226185: sandstone, Nicolay 1

(Nambeet Formation, Canning Basin)

Location and sampling

Joanna Spring (SF 51-3), McTavish (3556)
MGA Zone 51, 526953E 7825419N

Sampled on 18 February 2019

This sample was collected from the 3437.3 – 3438.1 m depth interval of drillcore Nicolay 1, from a petroleum well drilled in 2012 by New Standard Onshore Pty Ltd (New Standard Onshore, 2013). The drillhole is located on a north–south cleared line, about 52 km south of Battlement Rocks, 30 km southeast of Turkey Place Hill, and 18.8 km west-northwest of McTavish Claypan.

Geological context

The unit sampled is the Early Ordovician (485–470 Ma; Cohen et al. 2020) Nambeet Formation of the Canning Basin (Haines, 2004). The unit is about 337 m thick in Nicolay 1 and consists mainly of shale with interbedded limestone and sandstone (New Standard Onshore, 2013). The Nambeet Formation is the lowermost Canning Basin unit in Nicolay 1 and is conformably overlain by the Willara Formation (Haines, 2004; Zhan, 2019). Three additional samples of Nambeet Formation sandstone from Nicolay 1 yielded maximum depositional ages of 828–533 Ma (GSWA 226183, Wingate et al., 2021a; GSWA 226184, Wingate et al., 2021b; GSWA 226186, Wingate et al., 2021c). A Nambeet Formation sandstone sample from Samphire Marsh 1, drilled about 246 km to the west-northwest, yielded detrital zircon ages of 2838–506 Ma, and a conservative maximum depositional age of 519 ± 3 Ma (GSWA 135962, Wingate and Haines, 2009). Another Nambeet Formation sample from Barnicarndy 1, drilled about 185 km to the southwest, yielded detrital zircon ages of 3274–590 Ma, and a conservative maximum depositional age of 1007 ± 18 Ma (GSWA 237998, Wingate et al., 2021d).

Petrographic description

The sample is a fine-grained, fossiliferous sandstone, consisting of about 50% carbonate bioclasts, 40–45% quartz, 5% feldspar, 1% pyrite and accessory muscovite, tourmaline, glauconite and zircon. Carbonate bioclasts include fossil fragments, probable ooids, and small featureless carbonate grains. Fossil fragments include straight and curved, rod-like shell fragments, up to 5.5 mm long, some with scalloped outlines. The thin section contains two complete shells, one of which is up to 5 mm long, and is infilled with micritic carbonate and minor detrital quartz and feldspar grains; the shells and shell fragments are probable brachiopods. Slightly irregular carbonate plates up to 0.8 mm in size have finely pitted textures, and may be fragments of echinoid or coral. The sample also contains abundant circular and fewer oval probable ooids that are up to 0.2 mm diameter and exhibit radial textures. Quartz is mainly subangular, clear, unstrained, and up to 0.15 mm in diameter. Feldspar grains of the same size and shape as quartz are mostly unaltered plagioclase. Narrow laths of detrital muscovite are up to 0.15 mm long, and subrounded grains of gold–green tourmaline and green glauconite are up to 0.05 mm long. Pyrite is mainly irregular, and up to 0.1 mm diameter. In the least carbonate-cemented parts of the sandstone, quartz and feldspar form a closely packed, well sorted aggregate, which is texturally mature. The sandstone is from a marine environment and non-abraded shells suggest minimal transport. There is no evidence of deformation or metamorphism.

Zircon morphology

Zircons isolated from this sample are mostly colourless, anhedral to euhedral, and variably rounded. The crystals are up to 350 µm long, and equant to elongate, with aspect ratios up to 4:1. Many crystals have pitted outer surfaces and, in cathodoluminescence (CL) images, concentric zoning truncated at grain boundaries, features consistent with sedimentary transport. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 11 June 2020, using LA-ICP-MS instruments at the GeoHistory Facility in the John de Laeter Centre, Curtin University. Separated zircons mounted in epoxy disks were ablated using a Resonetics RESOlution M-50A-LR sampling system, incorporating a 193 nm Compex 102 excimer laser, and ion signal intensities were measured using an Agilent 8900s quadrupole ICP-MS. Samples were ablated for 22 s at a 5 Hz repetition rate, using a 24 µm beam diameter and laser energy of 2.0 J/cm^2. Zircon standard 91500 was used as the primary reference material, and GJ1, OGC1 and Plešovice were included as secondary standards. Additional analytical details are provided in Fielding et al. (2021). All relevant uncertainties have been propagated into the ratios and dates reported in Table 1. Ages from analyses for which $^{238}U/^{206}Pb$ ratios indicate dates <1000 Ma are based on 207-corrected $^{238}U/^{206}Pb^*$ ratios; those >1000 Ma are based on uncorrected $^{207}Pb/^{206}Pb$ ratios.

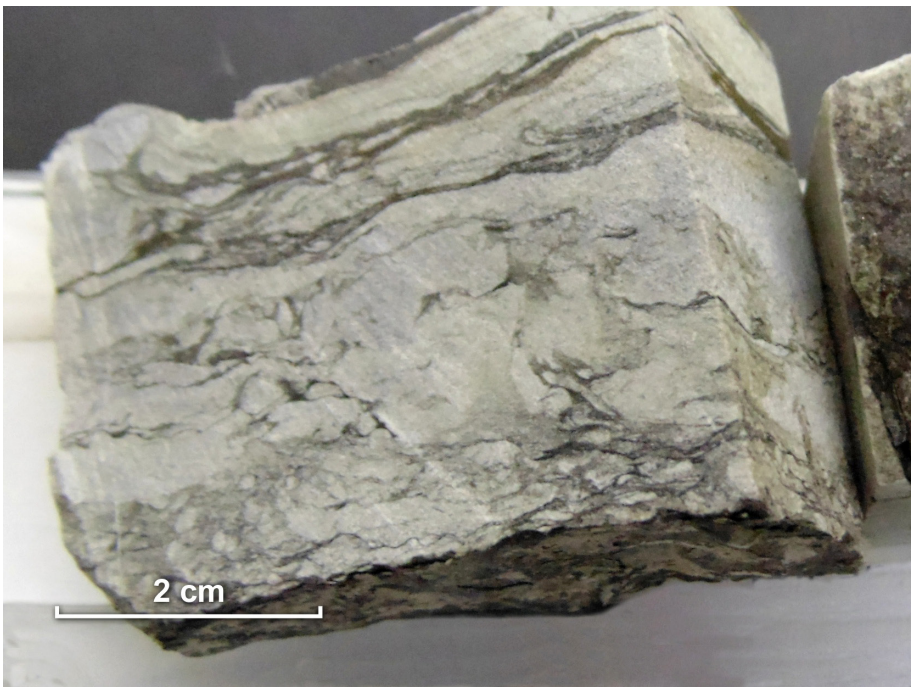

Figure 1. Drillcore image of sample 226185: sandstone, Nicolay 1

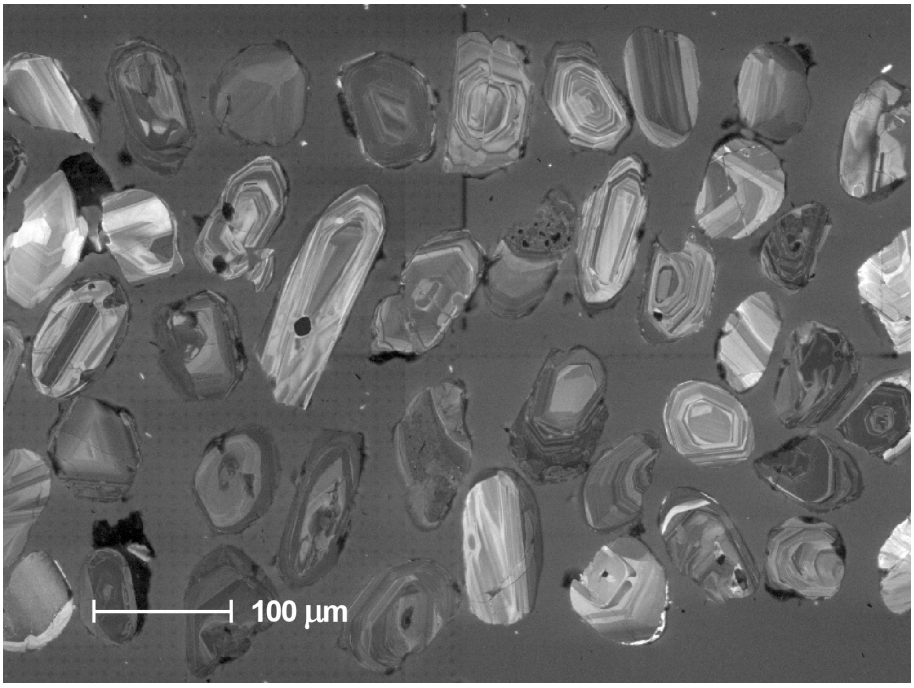

Figure 2. Cathodoluminescence image of representative zircons from sample 226185: sandstone, Nicolay 1

Table 1. LA-ICP-MS analytical results for zircons from sample 226185: sandstone, Nicolay 1

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		Disc. (%)
Y	71	71	398	345	0.87	13.312	0.078	0.05890	0.00105	467	3	563	39	17.1
S	44	44	207	167	0.81	13.053	0.083	0.05650	0.00130	476	3	472	51	-0.8
S	74	74	148	64	0.43	11.377	0.084	0.06040	0.00165	543	4	618	59	12.1
S	55	55	1620	1066	0.66	10.526	0.051	0.06146	0.00080	585	3	655	28	10.7
S	19	19	2406	108	0.04	10.503	0.047	0.05842	0.00070	586	3	546	26	-7.5
S	46	46	565	489	0.87	10.436	0.053	0.05910	0.00085	590	3	571	31	-3.3
S	58	58	2250	105	0.05	10.198	0.050	0.05990	0.00075	603	3	600	27	-0.5
S	22	22	824	1690	2.05	10.000	0.065	0.05980	0.00090	614	4	596	33	-3.0
S	35	35	468	572	1.22	9.867	0.054	0.06380	0.00105	622	3	735	35	15.3
S	41	41	86	112	1.29	9.833	0.087	0.06260	0.00185	624	5	695	63	10.1
S	21	21	133	69	0.52	9.524	0.077	0.05900	0.00130	644	5	567	48	-13.5
S	45	45	218	334	1.53	9.470	0.058	0.06210	0.00125	647	4	678	43	4.5
S	36	36	319	310	0.97	9.345	0.048	0.06150	0.00100	655	3	657	35	0.2
S	66	66	274	131	0.48	9.050	0.057	0.06150	0.00115	676	4	657	40	-2.9
S	24	24	358	76	0.21	8.828	0.047	0.06130	0.00105	692	3	650	37	-6.5
S	31	31	484	1450	3.00	7.112	0.114	0.07670	0.00105	848	13	1113	27	23.8
S	27	27	1531	87	0.06	6.916	0.053	0.07253	0.00085	871	6	1001	24	13.0
S	8	8	509	108	0.21	6.901	0.036	0.06640	0.00095	872	4	819	30	-6.5
S	80	80	465	225	0.48	6.684	0.051	0.07260	0.00100	899	6	1003	28	10.4
S	4	4	243	103	0.43	6.386	0.039	0.06880	0.00120	938	5	893	36	-5.1
S	7	7	431	167	0.39	5.931	0.030	0.07100	0.00100	1004	5	957	29	-4.9
S	18	18	369	282	0.76	6.035	0.031	0.07540	0.00110	988	5	1079	29	8.4
S	10	10	373	140	0.37	5.834	0.031	0.07250	0.00105	1020	5	1000	29	-2.0
S	47	47	237	152	0.64	5.747	0.031	0.07300	0.00125	1034	5	1014	35	-2.0
S	11	11	81	74	0.91	5.473	0.040	0.07390	0.00180	1082	7	1039	49	-4.1
S	17	17	415	268	0.64	5.402	0.026	0.07420	0.00095	1095	5	1047	26	-4.6
S	48	48	252	149	0.59	5.814	0.032	0.07420	0.00120	1023	5	1047	33	2.3
S	73	73	356	128	0.36	5.618	0.035	0.07550	0.00110	1056	6	1082	29	2.4

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	Disc. (%)
S	77	77	233	127	0.54	5.714	0.031	0.07590	0.00120	1040	5	1092	32	4.8
S	53	53	237	108	0.46	5.593	0.038	0.07610	0.00115	1060	7	1098	30	3.4
S	40	40	456	81	0.18	5.519	0.027	0.07630	0.00105	1074	5	1103	28	2.7
S	51	51	590	210	0.36	5.435	0.027	0.07690	0.00100	1089	5	1119	26	2.7
S	9	9	86	64	0.75	5.157	0.041	0.07750	0.00155	1142	8	1134	40	-0.7
S	76	76	75	55	0.73	5.126	0.041	0.07880	0.00180	1149	8	1167	45	1.6
S	52	52	147	159	1.09	5.149	0.028	0.07880	0.00140	1144	6	1167	35	2.0
S	1	1	362	215	0.59	4.931	0.024	0.07890	0.00115	1190	5	1170	29	-1.8
S	75	75	181	192	1.06	5.200	0.030	0.07980	0.00120	1134	6	1192	30	4.9
S	72	72	153	140	0.92	5.084	0.031	0.08020	0.00145	1158	6	1202	36	3.7
S	14	14	175	25	0.14	4.585	0.024	0.08090	0.00125	1272	6	1219	30	-4.3
S	32	32	86	93	1.08	4.869	0.036	0.08130	0.00160	1204	8	1229	39	2.0
S	26	26	125	229	1.83	4.613	0.028	0.08370	0.00150	1265	7	1286	35	1.6
S	33	33	933	428	0.46	4.419	0.027	0.08470	0.00105	1315	7	1309	24	-0.5
S	38	38	331	424	1.28	3.922	0.020	0.09120	0.00115	1464	7	1451	24	-0.9
S	20	20	120	129	1.08	3.877	0.025	0.09140	0.00155	1479	8	1455	32	-1.7
S	28	28	224	148	0.66	3.727	0.021	0.09350	0.00140	1532	8	1498	28	-2.3
S	15	15	386	535	1.39	3.646	0.019	0.09360	0.00120	1563	7	1500	24	-4.2
S	16	16	673	671	1.00	3.551	0.016	0.09626	0.00115	1599	6	1553	22	-3.0
S	64	64	154	119	0.77	3.706	0.020	0.09670	0.00155	1540	7	1561	30	1.4
S	59	59	1036	1634	1.58	3.580	0.019	0.09682	0.00110	1588	8	1564	21	-1.5
S	42	42	286	134	0.47	3.651	0.039	0.09810	0.00130	1561	15	1588	25	1.7
S	70	70	501	539	1.08	3.635	0.019	0.09830	0.00120	1567	7	1592	23	1.6
S	81	81	91	127	1.40	3.682	0.024	0.09860	0.00170	1549	9	1598	32	3.1
S	54	54	104	95	0.91	3.656	0.022	0.09890	0.00155	1559	8	1603	29	2.8
S	30	30	454	85	0.19	3.264	0.016	0.10760	0.00130	1723	7	1759	22	2.1
S	56	56	212	179	0.85	2.902	0.015	0.11810	0.00150	1909	8	1928	23	1.0

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	Disc. (%)
D	68	68	5060	4129	0.82	14.368	0.124	0.08490	0.00225	434	4	1313	51	67.0
D	37	37	549	149	0.27	5.417	0.026	0.07370	0.00095	1092	5	1033	26	-5.7
D	6	6	109	118	1.08	5.048	0.032	0.07390	0.00150	1165	7	1039	41	-12.2
D	29	29	1610	198	0.12	5.956	0.035	0.07459	0.00085	1001	6	1057	23	5.4
D	2	2	21	6	0.30	4.965	0.055	0.07490	0.00310	1183	12	1066	83	-11.0
D	39	39	665	170	0.26	4.953	0.031	0.07720	0.00095	1186	7	1126	25	-5.3
D	57	57	138	91	0.66	5.260	0.035	0.08020	0.00130	1122	7	1202	32	6.7
D	65	65	206	251	1.22	5.896	0.035	0.08140	0.00130	1010	6	1231	31	18.0
D	78	78	360	962	2.67	5.063	0.031	0.08180	0.00115	1162	6	1241	28	6.4
D	49	49	136	111	0.82	5.408	0.029	0.08200	0.00145	1094	5	1246	35	12.2
D	62	62	256	161	0.63	4.924	0.030	0.08520	0.00130	1192	7	1320	30	9.7
D	63	63	106	72	0.68	4.589	0.031	0.08610	0.00170	1271	8	1340	38	5.2
D	67	67	346	123	0.36	4.191	0.025	0.09220	0.00120	1379	7	1472	25	6.3
D	60	60	117	123	1.05	4.175	0.034	0.09400	0.00160	1384	10	1508	32	8.2
D	25	25	425	289	0.68	3.978	0.036	0.09510	0.00125	1446	12	1530	25	5.5
D	69	69	346	182	0.53	4.211	0.022	0.09590	0.00125	1374	7	1546	24	11.1
D	61	61	412	391	0.95	4.227	0.027	0.10080	0.00125	1369	8	1639	23	16.5
D	23	23	1212	484	0.40	3.396	0.021	0.11000	0.00130	1664	9	1799	22	7.5
D	50	50	236	253	1.07	3.233	0.016	0.11220	0.00145	1737	7	1835	23	5.3
D	79	79	409	403	0.99	3.310	0.018	0.11480	0.00140	1702	8	1877	22	9.3
D	34	34	409	148	0.36	2.946	0.015	0.12620	0.00150	1884	8	2046	21	7.9
D	12	12	151	158	1.05	2.723	0.015	0.13210	0.00180	2016	9	2126	24	5.2
D	5	5	127	47	0.37	2.476	0.014	0.16670	0.00215	2187	10	2525	22	13.4
D	43	43	419	23	0.06	1.915	0.009	0.20750	0.00235	2708	10	2886	18	6.2
D	3	3	184	138	0.75	2.007	0.014	0.25500	0.00300	2606	15	3216	19	19.0
D	13	13	1351	678	0.50	3.546	0.126	0.26000	0.01650	1601	50	3247	100	50.7

Results

Eighty-one analyses were obtained from 81 zircons. Results are listed in Table 1, and shown in a concordia diagrams (Fig. 3) and probability density plots (Figs 4, 5).

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Twenty-five analyses >1000 Ma are >5% discordant, and one analysis <1000 Ma indicates high common Pb (*f*207 >2%). The dates obtained from these 26 analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 55 analyses can be divided into two groups, based on their $^{207}Pb/^{206}Pb$ and $^{238}U/^{206}Pb^*$ ratios.

Group Y comprises one analysis (Table 1), which yields a 207-corrected $^{238}U/^{206}Pb^*$ date of 466 ± 3 Ma (1σ).

Group S comprises 54 analyses (Table 1), which yield $^{207}Pb/^{206}Pb$ or 207-corrected $^{238}U/^{206}Pb^*$ dates of 1928–476 Ma.

It is possible that all of the analyses are of unmodified detrital zircons, in which case the date of 466 ± 3 Ma (1σ) for the single analysis in Group Y represents a maximum depositional age for the sandstone. A more conservative estimate of the maximum depositional age can be based on the weighted mean 207-corrected $^{238}U/^{206}Pb^*$ date of 589 ± 8 Ma (MSWD = 1.3) for the youngest coherent group of three analyses in Group S.

The data for combined Groups Y and S indicate significant age components at c. 1576, 1489, 1299, 1183, 1100, 985, 654, 619, and 587 Ma, based on contributions from approximately 7, 4, 3, 8, 10, 6, 3, 3, and 3 analyses, respectively (Fig. 5). These are interpreted as the ages of zircon-crystallizing rocks in the detrital source region(s), or as the ages of detrital components within sediments that have been reworked into this rock.

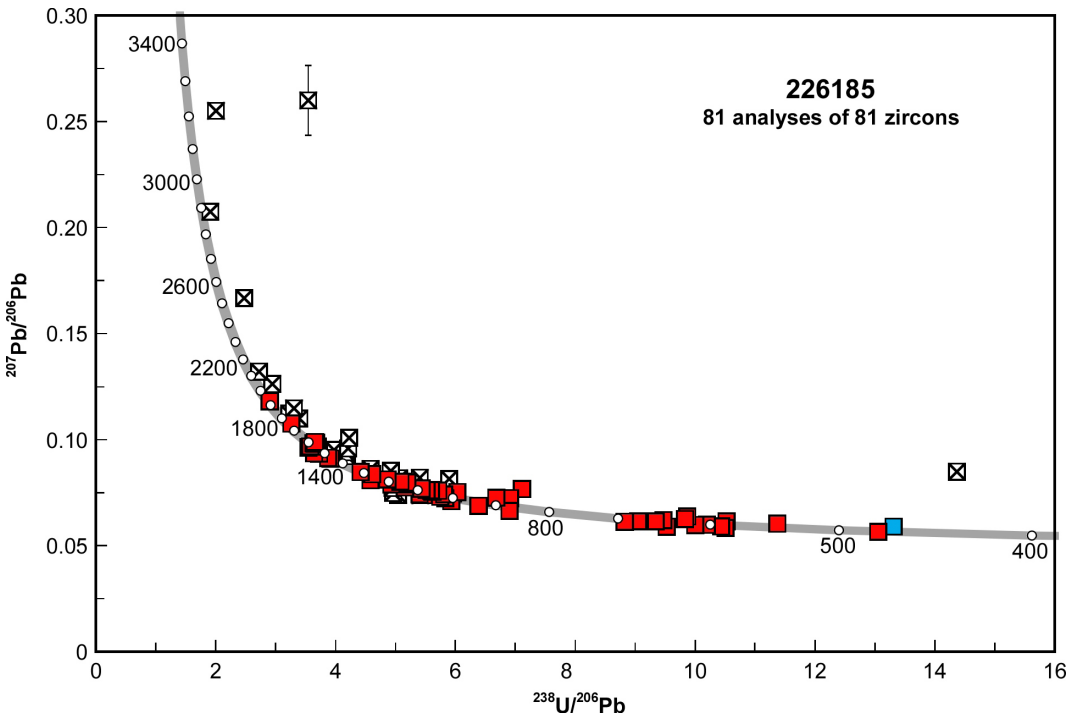

Figure 3. U–Pb analytical data, not corrected for common Pb, for zircons from sample 226185: sandstone, Nicolay 1. Blue square indicates Group Y (youngest detrital zircon); red squares indicate Group S (older detrital zircons); crossed squares indicate Group D (date >1000 Ma and discordance >5% or date <1000 Ma and f_{207} >2%)

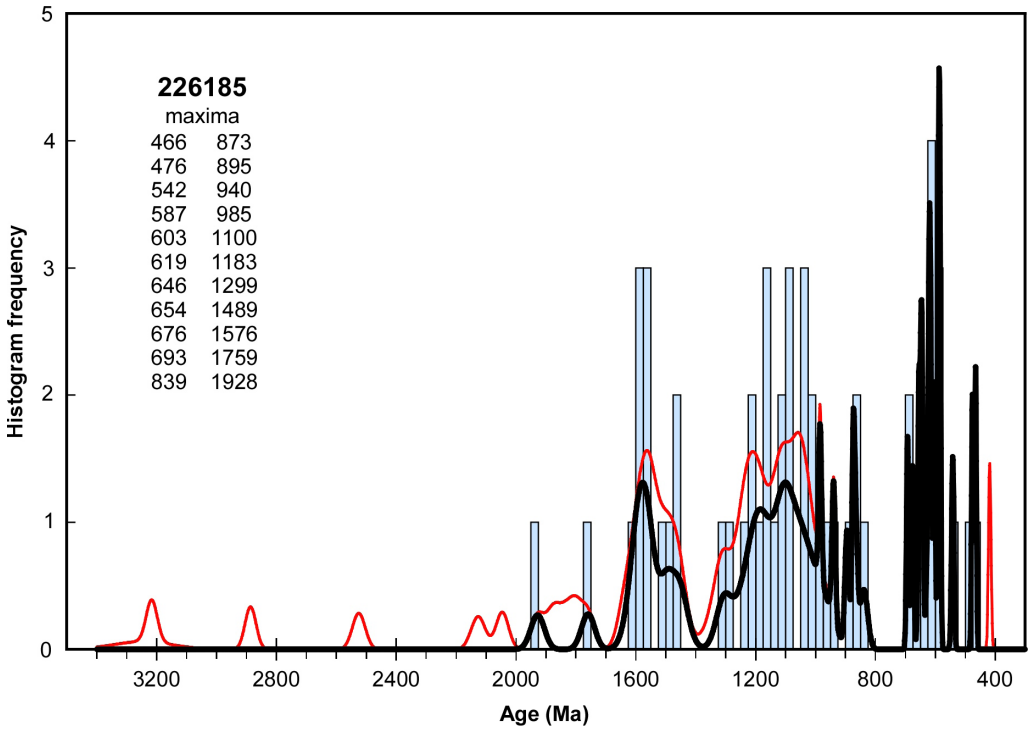

Figure 4. Probability density diagram and histogram for sample 226185: sandstone, Nicolay 1. Thick curve, maxima values, and frequency histogram (bin width 25 Ma) only include accepted data (55 analyses of 55 zircons). Thin curve includes all data (81 analyses of 81 zircons)

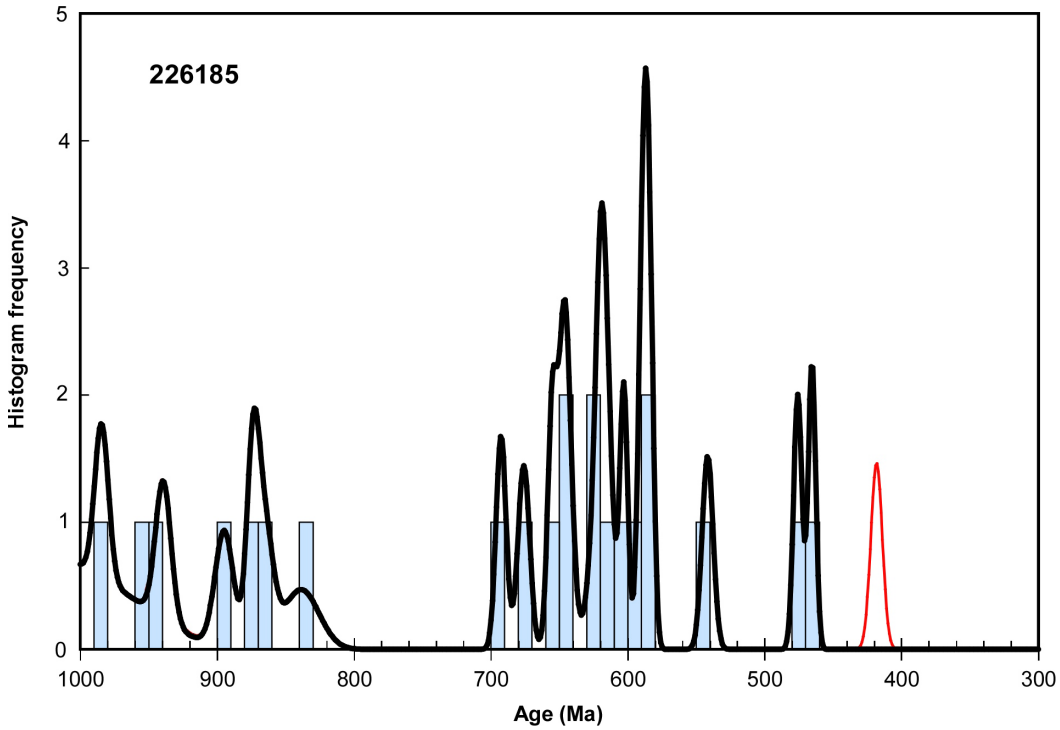

Figure 5. Expanded view of probability density diagram and histogram for sample 226185: sandstone, Nicolay 1. Bin width is 10 Ma

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Recommended reference for this publication

Wingate, MTD, Lu, Y and Dent, LM 2021, 226185: sandstone, Nicolay 1; Geochronology Record 1762: Geological Survey of Western Australia, 8p.

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